INVERTIS UNIVERSITY BAREILLY

Established by Govt. of U.P. u/s 2f of UGC Act, 1956 vide U.P. Act 22 of 2010.

BCS-403	Operating Systems	L T P 3 0 0	3 credits

Course Objectives:

CO1	To understand the main components of an OS & their functions.
CO2	To study the process management and scheduling.
CO3	To understand various issues in Inter Process Communication (IPC) and the role of OS in IPC.
CO4	To understand the concepts and implementation Memory management policies and virtual memory.
CO5	To understand the concept of file organization and access mechanism.

MODULE-I

Introduction: Concept of Operating Systems, Generations of Operating systems, Types of Operating Systems, OS Services, System Calls, Structure of an OS - Layered, Monolithic, Microkernel Operating Systems, Concept of Virtual Machine. Case study on UNIX and WINDOWS Operating System.

Processes: Definition, Process Relationship, Different states of a Process, Process State transitions, Process Control Block (PCB), Context switching.
Thread: Definition, Various states, Benefits of threads, Types of threads, Concept of multithreads.
Process Scheduling: Foundation and Scheduling objectives, Types of Schedulers, **Scheduling criteria:** CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time; Scheduling algorithms: Pre-emptive and Non pre-emptive, FCFS, SJF, RR; Multiprocessor scheduling.

MODULE-II

Inter-process Communication: Critical Section, Race Conditions, Mutual Exclusion, Hardware Solution, Dekker's solution, Peterson's Solution, The Producer/Consumer Problem, Semaphores, Event Counters, Monitors, Message Passing, Classical IPC Problems: Reader's & Writer Problem, Dinning Philosopher Problem etc.
Deadlocks: Definition, Necessary and sufficient conditions for Deadlock, Deadlock Prevention, Deadlock Avoidance: Banker's algorithm, Deadlock detection and Recovery.

MODULE-III

Memory Management: Basic concept, Logical and Physical address map, Memory allocation: Contiguous Memory allocation – Fixed and variable partition– Internal and External fragmentation and Compaction; Paging: Page fault, Demand paging, Page Replacement algorithms: Optimal, First in First Out (FIFO), Second Chance (SC), Not recently used (NRU) and Least Recently used (LRU).
Virtual Memory: Basics of Virtual Memory – Hardware and control structures – Locality of reference
I/O Hardware: I/O devices, Device controllers, Direct memory access Principles of I/O Software: Goals of Interrupt handlers, Device drivers.
File Management: Concept of File, Access methods, File types, File operation, Directory structure, Allocation methods (contiguous, linked, indexed), Free-space management (bit vector, linked list, grouping).
Disk Management: Disk structure, Disk scheduling - FCFS, SSTF, SCAN, C-SCAN, Disk reliability, Disk formatting, Boot-block, Bad blocks.

Text Books:

1. Silberschatz, Galvin and Gagne, "Operating Systems Concepts", Wiley
2. Sibsankar Halder and Alex A Aravind, "Operating Systems", Pearson Education
3. Harvey M Dietel, " An Introduction to Operating System", Pearson Education
4. D M Dhamdhere, "Operating Systems : A Concept based Approach", 2nd Edition, 13 TMH
5. William Stallings, "Operating Systems: Internals and Design Principles", 6th Edition, Pearson Education.

Course Outcomes:

CO1	Describe the important computer system resources and the role of operating system in their management policies and algorithms.
CO2	Understand the process management policies and scheduling of processes by CPU.
CO3	Evaluate the requirement for process synchronization and coordination handled by operating system.
CO4	Describe and analyze the memory management and its allocation policies.
CO5	Set file access permissions and protect and secure files.